Dolapo Jolade Bello AMRSC

Synthetic Organic and Medicinal Chemist | PhD Researcher, Organic Chemistry

Loughborough, UK | +44 7901 036402 | dolapobello2013@gmail.com

Portfolio: dolapobello.github.io | **LinkedIn:** linkedin.com/in/dolapo-bello

PROFILE

Synthetic organic chemist with a published record in medicinal chemistry, undertaking a fully funded PhD at Loughborough University on bicyclo[1.1.1]pentane (BCP) analogues of quinolone antibiotics. Designs and runs multistep routes to novel three dimensional bioisosteres, optimises difficult strained ring chemistry, and fully characterises new compounds by NMR, HRMS and HPLC. Three research degrees across organic, physical and materials chemistry give a broad, adaptable lab skill set. Award winning communicator who explains complex science clearly to specialist and general audiences. Seeking a Medicinal Chemist, Synthetic Chemist or R&D Scientist role in pharmaceutical, agrochemical or chemical industry research.

CORE COMPETENCIES

Medicinal chemistry | Multistep organic synthesis | Bioisosteric design | SAR and SPR analysis | Reaction optimisation | Heterocyclic chemistry | Strained ring chemistry | Air and moisture sensitive techniques | NMR, HRMS, HPLC, GC | Purification and chromatography | Scientific communication | Mentoring

SELECTED ACHIEVEMENTS

- **2 peer reviewed publications**, including The Journal of Organic Chemistry (2025), with a further manuscript in preparation.
- **3 presentation prizes in 2026**: Three Minute Thesis Winner and People's Choice (Loughborough University) and RSC Advances 5MT Runner Up (Women in Chemistry Conference, University of Bristol).
- **Selected for RSC Broadening Horizons 2026**, a year long industry programme with GSK, AstraZeneca, Unilever, Johnson Matthey and Syngenta.
- **Associate Member of the Royal Society of Chemistry** (AMRSC), admitted June 2026.

RESEARCH EXPERIENCE

PhD Researcher, Organic and Medicinal Chemistry | Loughborough University **2024 – Present**

Fully funded studentship. Thesis: Bicyclo[1.1.1]pentane Analogues of Quinolone Antibiotics. Supervisor: Dr Marc Kimber

- Designed and synthesised BCP containing quinolone analogues as three dimensional replacements for flat aromatic rings, targeting new antibiotics against resistant bacteria.
- Developing a route to N-BCP quinolin-4(1H)-ones through masked quinolinium intermediates and extending it to fluorinated and 3-aryl substituted analogues for SAR studies.
- Co-authored a J. Org. Chem. (2025) paper reporting nucleophilic substitution of 1,3-diiodobicyclo[1.1.1]pentane to make BCP pyridinium, quinolinium, isoquinolinium and pyrazolium salts.
- Optimised reaction conditions and explored substrate scope and mechanism to improve yield, efficiency and selectivity of strained ring couplings.
- Characterised every new compound by 1D/2D NMR, HRMS/MS, IR, HPLC and GC; handled air and moisture sensitive reagents using Schlenk line and glovebox techniques.
- Demonstrate in undergraduate organic chemistry teaching labs and support MSc research students with experimental design and data analysis.

MSc Researcher, Chemistry | Nottingham Trent University **2023 – 2024**

- Designed and analysed microfluidic T junction systems to measure liquid mixing, diffusion and flow, using quantitative analysis to optimise mixing performance. Supervisor: Dr Cheu Wing Li.

MSc Researcher, Industrial Chemistry | Obafemi Awolowo University **2018 – 2020**

- Synthesised and characterised oxidised starch nanoparticles that remove toxic lead and cadmium from wastewater; modelled adsorption with kinetic, thermodynamic and isotherm models. Published 2021.

Undergraduate Researcher, Industrial Chemistry | Obafemi Awolowo University **2009 – 2015**

- Characterised Nigerian and Canadian tar sands and commercial materials using AAS, ICP-MS, FTIR, SEM and XRD.

PUBLICATIONS

- Monroe, H.J.C.; **Bello, D.J.**; Duff, B.J.; Elsegood, M.R.J.; Watanabe, K.; Pritchard, G.J.; Kimber, M.C. Nucleophilic Substitution of 1,3-Diiodobicyclo[1.1.1]pentane: Synthesis of Bicyclo[1.1.1]pentylpyridinium, Quinolinium, Isoquinolinium, and Pyrazolium Salts. *J. Org. Chem.* 2025, 90 (23), 7712–7722. DOI: 10.1021/acs.joc.5c00565
- Awokoya, K.N.; Oninla, V.O.; **Bello, D.J.** Synthesis of oxidised starch (Dioscorea dumetorum) nanoparticles for the adsorption of lead(II) and cadmium(II) ions from wastewater. *Environ. Nanotechnol. Monit. Manag.* 2021, 14, 100440. DOI: 10.1016/j.enmm.2021.100440
- **Bello, D.J.** et al. N-Bicyclo[1.1.1]pentylation of quinolin-4(1H)-ones via masked quinolinium intermediates: access to fluorinated and 3-aryl substituted analogues. *Manuscript in preparation.*

EDUCATION

PhD Chemistry (Organic Chemistry) | Loughborough University, UK — **2024 – Present**

MSc Chemistry | Nottingham Trent University, UK — **2023 – 2024**

MSc Industrial Chemistry | Obafemi Awolowo University, Nigeria — **2018 – 2020**

BSc Industrial Chemistry | Obafemi Awolowo University, Nigeria — **2009 – 2015**

TECHNICAL SKILLS

Synthesis: Multistep synthesis, retrosynthesis, reaction optimisation, C–C and C–N bond formation, cross coupling, heterocycle and strained ring (BCP) chemistry, functional group transformations

Medicinal chemistry: Bioisosteric replacement, molecular design, SAR and structure property relationships, antibiotic drug discovery

Analytical: 1D/2D NMR, HRMS/MS, IR, HPLC, GC, UV–Vis, XRD, SEM, TEM, ICP, AAS

Laboratory: Schlenk line, glovebox, inert atmosphere synthesis, flash chromatography, purification, rotary evaporation

Software: ChemDraw, Chem3D, MestReNova, TopSpin, Mercury (CCDC), OPUS/OMNIC, EndNote, Mendeley

INDUSTRY ENGAGEMENT AND PRESENTATIONS

- RSC Broadening Horizons in the Chemical Sciences, 5th cohort (2026): taster event in Manchester and site visit to Unilever, Port Sunlight; partners include GSK, AstraZeneca, Unilever, Johnson Matthey, Syngenta, AM Technology and Nanomerics.
- Oral presentation, Science Matters Conference, Loughborough University, June 2026.
- Five Minute Thesis talk, Celebrating Women in Chemistry Conference, University of Bristol, May 2026.
- Delegate, RSC Heterocyclic and Synthesis Group 3rd Postdoctoral Symposium (virtual), May 2026.
- Poster presentations at RSC organic chemistry meetings, 2024 – 2026, including University of Sheffield, April 2026.

LEADERSHIP AND OUTREACH

- Women in Science (Nigeria): helps organise the annual February event celebrating women scientists and encouraging girls into science.
- Peer Educator, UNICEF Nigeria: delivered health awareness sessions to secondary school students.
- Conference Coordinator, International Institute for Sustainable Development, 2016; Financial Secretary (student executive), Department of Chemistry, Obafemi Awolowo University, 2014.

PROFESSIONAL MEMBERSHIPS

Associate Member, Royal Society of Chemistry (AMRSC), 2026 | Black Women in Science (UK), 2024 – Present | Women in Science (Nigeria), 2021 – Present

Full research summary, publications and certificates: dolapobello.github.io. *References available on request.*